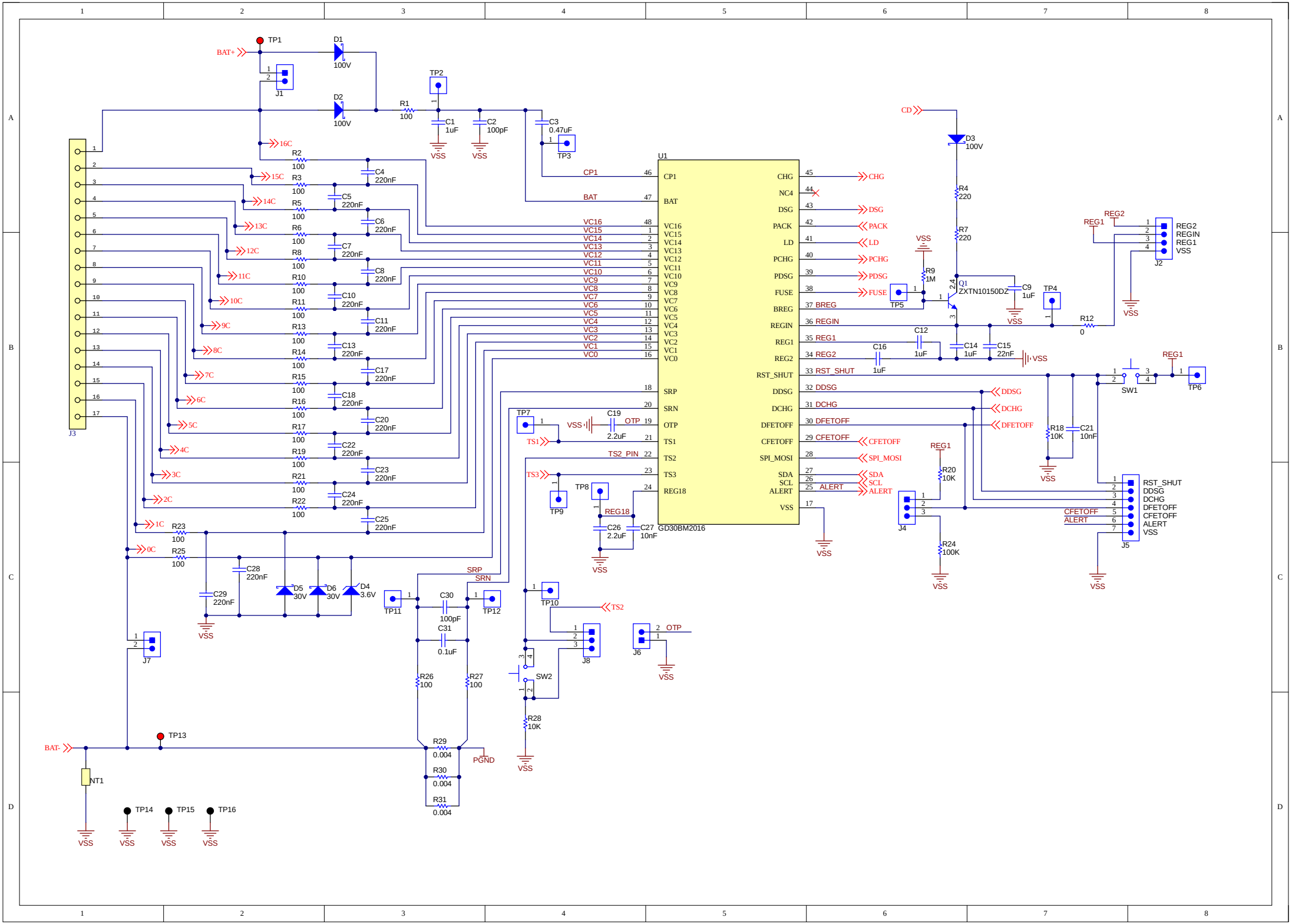
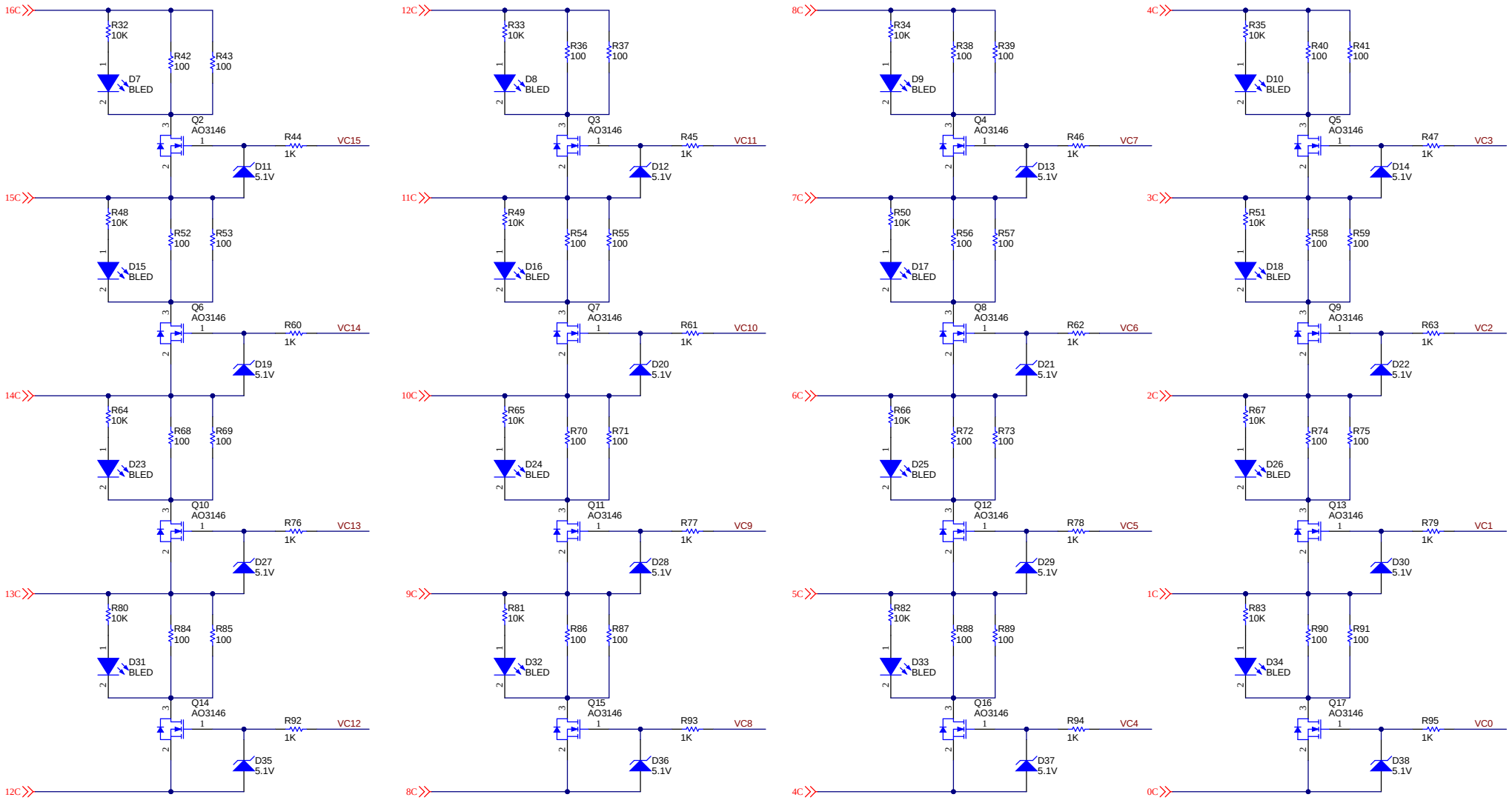
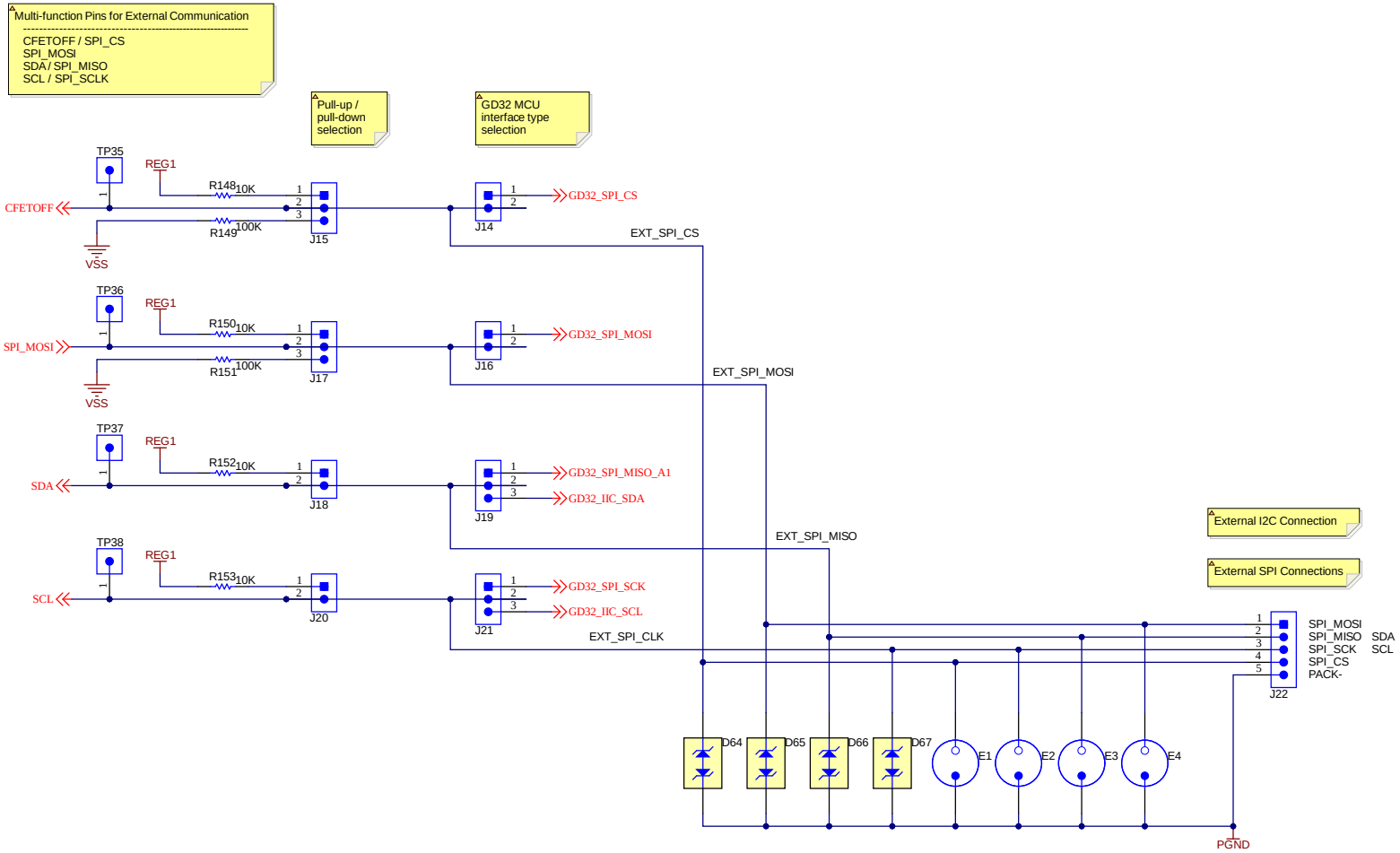
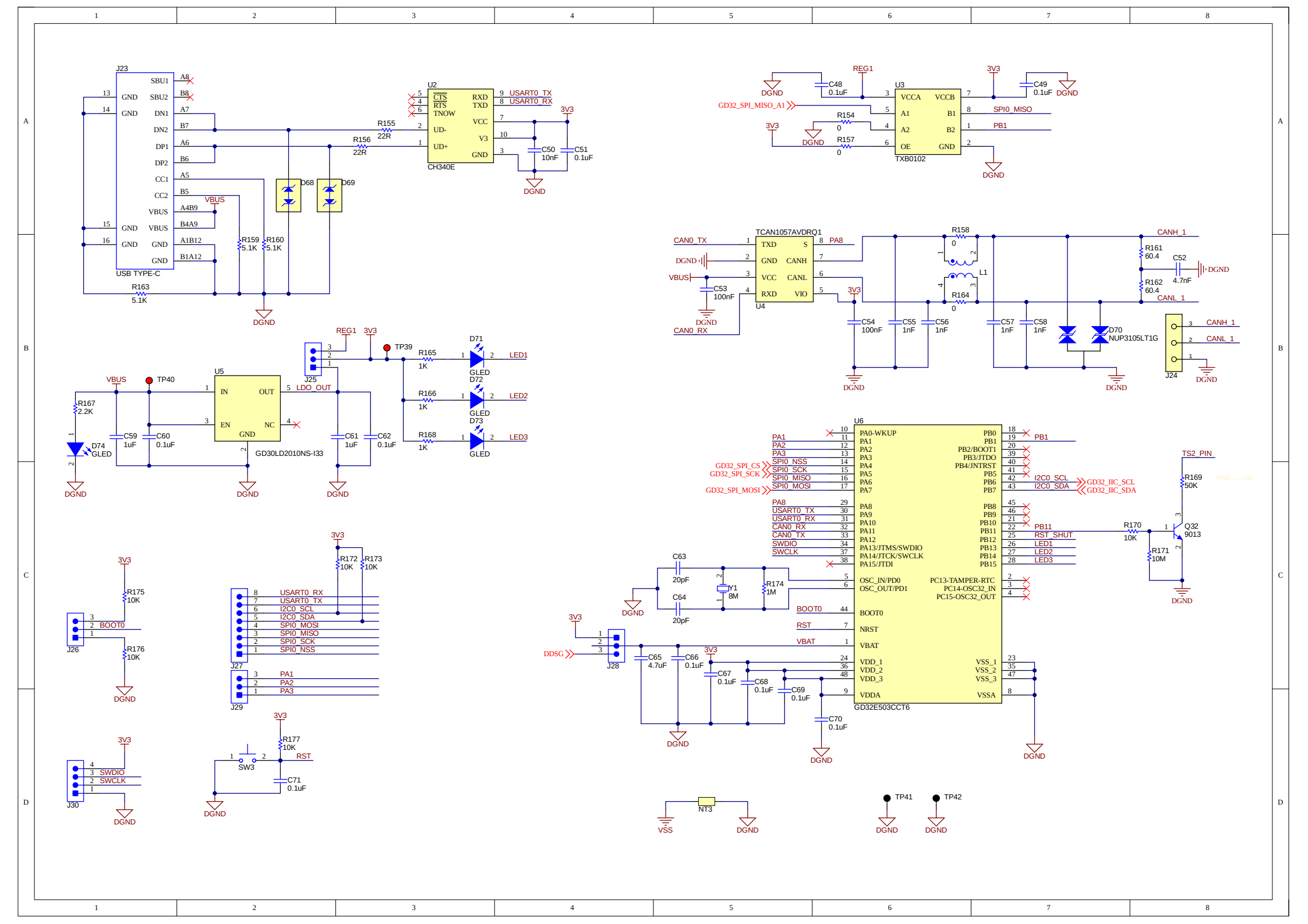
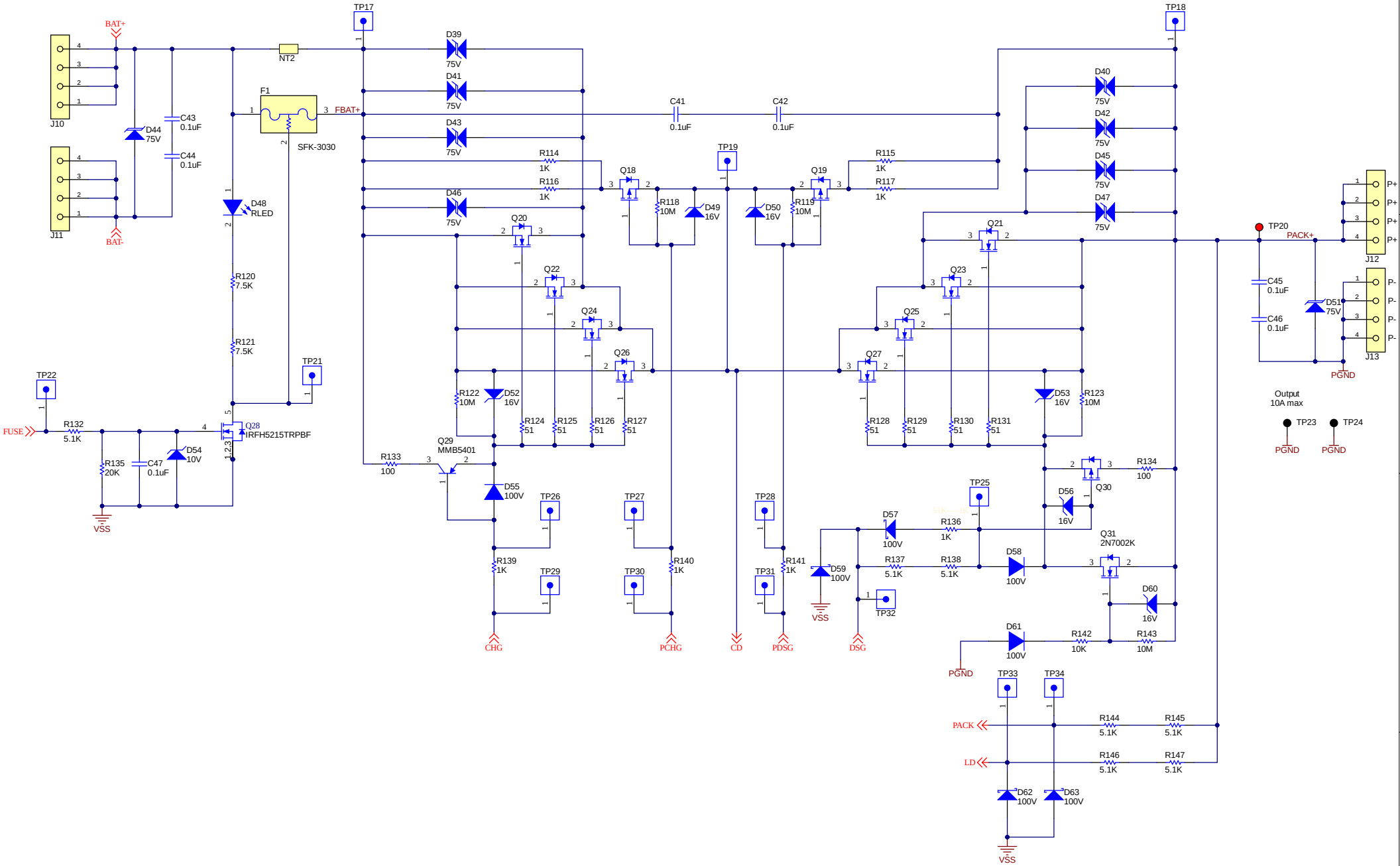
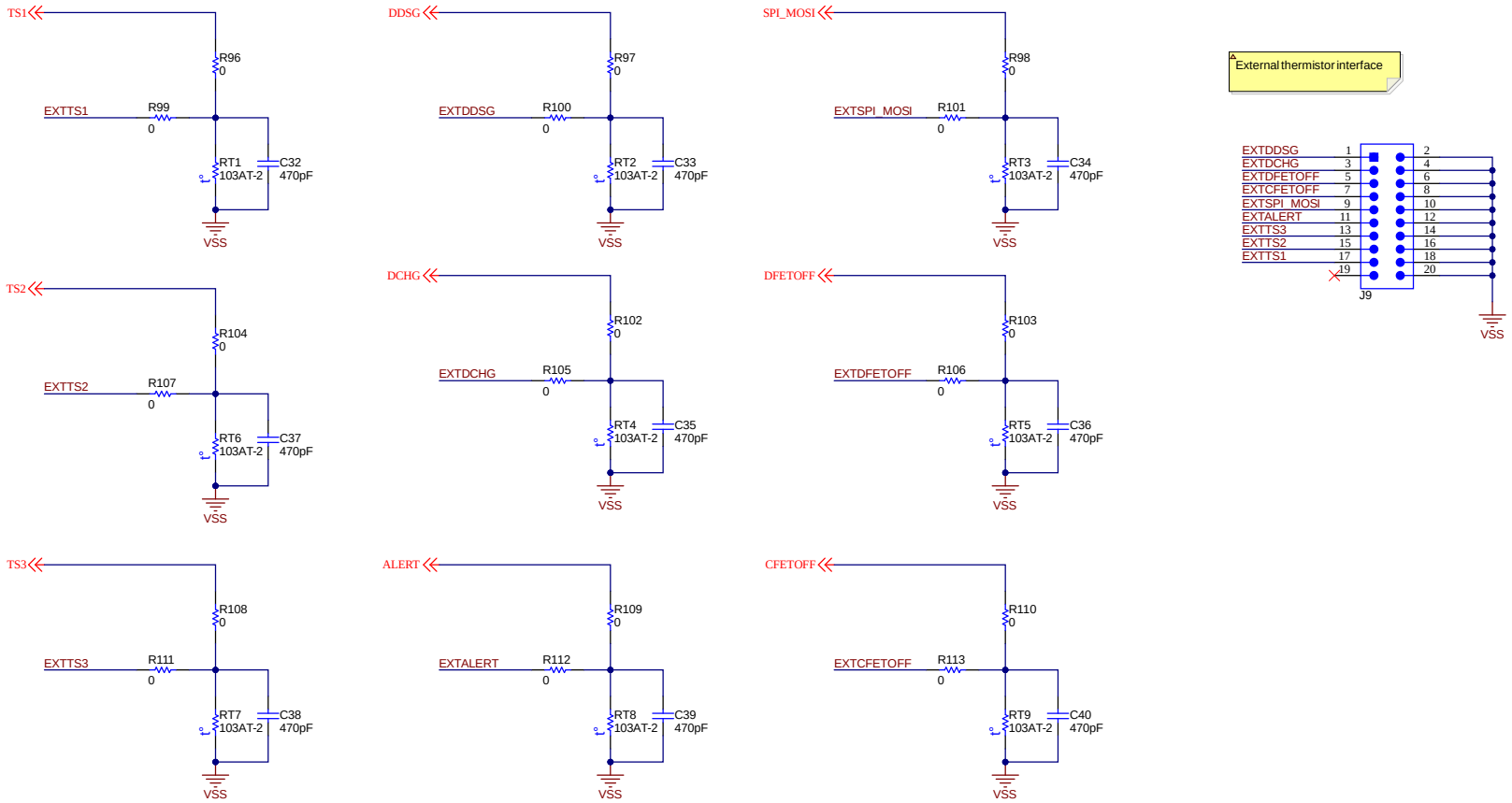

U1
GD30BM2016
BAT+
BAT-
J1
J2
J3
J4
J5
J6
J7
J8
D1
D2
D3
D4
D5
D6
100V
30V
3.6V
Q1
ZXTN10150DZ
SW1
SW2
NT1
TP1
TP2
TP3
TP4
TP5
TP6
TP7
TP8
TP9
TP10
TP11
TP12
TP13
TP14
TP15
TP16
R1 100
R2 100
R3 100
R4 220
R5 100
R6 100
R7 220
R8 100
R9 1M
R10 100
R11 100
R12 0
R13 100
R14 100
R15 100
R16 100
R17 100
R18 10K
R19 100
R20 10K
R21 100
R22 100
R23 100
R24 100K
R25 100
R26 100
R27 100
R28 10K
R29 0.004
R30 0.004
R31 0.004
C1 1uF
C2 100pF
C3 0.47uF
C4 220nF
C5 220nF
C6 220nF
C7 220nF
C8 220nF
C9 1uF
C10 220nF
C11 220nF
C12 1uF
C13 220nF
C14 1uF
C15 22nF
C16 1uF
C17 220nF
C18 220nF
C19 2.2uF
C20 220nF
C21 10nF
C22 220nF
C23 220nF
C24 220nF
C25 220nF
C26 2.2uF
C27 10nF
C28 220nF
C29 220nF
C30 100pF
C31 0.1uF
CP1
BAT
VC16
VC15
VC14
VC13
VC12
VC11
VC10
VC9
VC8
VC7
VC6
VC5
VC4
VC3
VC2
VC1
VC0
SRP
SRN
OTP
TS1
TS2_PIN
TS3
REG18
CHG
NC4
DSG
PACK
LD
PCHG
PDSG
FUSE
BREG
REGIN
REG1
REG2
RST_SHUT
DDSG
DCHG
DFETOFF
CFETOFF
SPI_MOSI
SDA
SCL
ALERT
VSS
PGND
CD
REG2
REGIN
REG1
VSS
RST_SHUT
DDSG
DCHG
DFETOFF
CFETOFF
ALERT
VSS
16C
15C
14C
13C
12C
11C
10C
9C
8C
7C
6C
5C
4C
3C
2C
1C
0C


16C
R32 10K
R42 100
R43 100
D7 BLED
Q2 AO3146
R44 1K
VC15
D11 5.1V
15C
R48 10K
R52 100
R53 100
D15 BLED
Q6 AO3146
R60 1K
VC14
D19 5.1V
14C
R64 10K
R68 100
R69 100
D23 BLED
Q10 AO3146
R76 1K
VC13
D27 5.1V
13C
R80 10K
R84 100
R85 100
D31 BLED
Q14 AO3146
R92 1K
VC12
D35 5.1V
12C
12C
R33 10K
R36 100
R37 100
D8 BLED
Q3 AO3146
R45 1K
VC11
D12 5.1V
11C
R49 10K
R54 100
R55 100
D16 BLED
Q7 AO3146
R61 1K
VC10
D20 5.1V
10C
R65 10K
R70 100
R71 100
D24 BLED
Q11 AO3146
R77 1K
VC9
D28 5.1V
9C
R81 10K
R86 100
R87 100
D32 BLED
Q15 AO3146
R93 1K
VC8
D36 5.1V
8C
8C
R34 10K
R38 100
R39 100
D9 BLED
Q4 AO3146
R46 1K
VC7
D13 5.1V
7C
R50 10K
R56 100
R57 100
D17 BLED
Q8 AO3146
R62 1K
VC6
D21 5.1V
6C
R66 10K
R72 100
R73 100
D25 BLED
Q12 AO3146
R78 1K
VC5
D29 5.1V
5C
R82 10K
R88 100
R89 100
D33 BLED
Q16 AO3146
R94 1K
VC4
D37 5.1V
4C
4C
R35 10K
R40 100
R41 100
D10 BLED
Q5 AO3146
R47 1K
VC3
D14 5.1V
3C
R51 10K
R58 100
R59 100
D18 BLED
Q9 AO3146
R63 1K
VC2
D22 5.1V
2C
R67 10K
R74 100
R75 100
D26 BLED
Q13 AO3146
R79 1K
VC1
D30 5.1V
1C
R83 10K
R90 100
R91 100
D34 BLED
Q17 AO3146
R95 1K
VC0
D38 5.1V
0C


Multi-function Pins for External Communication
CFETOFF / SPI_CS
SPI_MOSI
SDA / SPI_MISO
SCL / SPI_SCLK
Pull-up / pull-down selection
GD32 MCU interface type selection
TP35
REG1
R148 10K
R149 100K
J15
CFETOFF
VSS
J14
GD32_SPI_CS
EXT_SPI_CS
TP36
R150 10K
R151 100K
J17
SPI_MOSI
J16
GD32_SPI_MOSI
EXT_SPI_MOSI
TP37
R152 10K
J18
SDA
J19
GD32_SPI_MISO_A1
GD32_IIC_SDA
EXT_SPI_MISO
TP38
R153 10K
J20
SCL
J21
GD32_SPI_SCK
GD32_IIC_SCL
EXT_SPI_CLK
External I2C Connection
External SPI Connections
J22
SPI_MOSI
SPI_MISO SDA
SPI_SCK SCL
SPI_CS
PACK-
D64
D65
D66
D67
E1
E2
E3
E4
PGND


J23
USB TYPE-C
U2
CH340E
U3
TXB0102
REG1
3V3
DGND
VBUS
R155 22R
R156 22R
R159 5.1K
R160 5.1K
R163 5.1K
D68
D69
C48 0.1uF
C49 0.1uF
C50 10nF
C51 0.1uF
GD32_SPI_MISO_A1
SPI0_MISO
PB1
R154 0
R157 0
U4
TCAN1057AVDRQ1
CAN0_TX
CAN0_RX
C53 100nF
R158 0
L1
R164 0
C54 100nF
C55 1nF
C56 1nF
C57 1nF
C58 1nF
D70 NUP3105LT1G
CANH_1
CANL_1
R161 60.4
R162 60.4
C52 4.7nF
J24
U5
TP40
TP39
J25
LDO_OUT
R165 1K
R166 1K
R168 1K
D71 GLED
D72 GLED
D73 GLED
LED1
LED2
LED3
R167 2.2K
D74 GLED
C59 1uF
C60 0.1uF
C61 1uF
C62 0.1uF
GD30LD2010NS-I33
U6
GD32E503CCT6
PA0-WKUP
PB1
I2C0_SCL
I2C0_SDA
GD32_IIC_SCL
GD32_IIC_SDA
GD32_SPI_CS
GD32_SPI_SCK
GD32_SPI_MOSI
SPI0_NSS
SPI0_SCK
SPI0_MISO
SPI0_MOSI
USART0_TX
USART0_RX
CAN0_RX
CAN0_TX
SWDIO
SWCLK
PB11
RST_SHUT
LED1
LED2
LED3
TS2_PIN
R169 50K
R170 10K
R171 10M
Q32 9013
C63 20pF
C64 20pF
Y1 8M
R174 1M
BOOT0
RST
VBAT
DDSG
J28
C65 4.7uF
C66 0.1uF
C67 0.1uF
C68 0.1uF
C69 0.1uF
C70 0.1uF
J26
R175 10K
R176 10K
J27
R172 10K
R173 10K
J29
J30
SWDIO
SWCLK
SW3
R177 10K
C71 0.1uF
NT3
VSS
TP41
TP42


BAT+
BAT-
J10
J11
C43 0.1uF
C44 0.1uF
D44 75V
NT2
F1
SFK-3030
FBAT+
D48 RLED
R120 7.5K
R121 7.5K
TP17
TP18
TP19
TP20
TP21
TP22
D39 75V
D41 75V
D43 75V
D46 75V
D40 75V
D42 75V
D45 75V
D47 75V
C41 0.1uF
C42 0.1uF
R114 1K
R116 1K
R115 1K
R117 1K
Q18
Q19
Q20
Q21
Q22
Q23
Q24
Q25
Q26
Q27
R118 10M
D49 16V
D50 16V
R119 10M
PACK+
C45 0.1uF
C46 0.1uF
D51 75V
J12
J13
P+
P-
PGND
Output
10A max
TP23
TP24
FUSE
R132 5.1K
R135 20K
C47 0.1uF
D54 10V
Q28 IRFH5215TRPBF
VSS
R122 10M
D52 16V
R124 51
R125 51
R126 51
R127 51
R128 51
R129 51
R130 51
R131 51
D53 16V
R123 10M
Q29 MMB5401
R133 100
D55 100V
TP26
TP27
TP28
R139 1K
R140 1K
R141 1K
TP29
TP30
TP31
CHG
PCHG
CD
PDSG
DSG
TP25
TP32
D57 100V
R136 1K
R137 5.1K
R138 5.1K
D59 100V
D58 100V
Q30
Q31 2N7002K
R134 100
D56 16V
D60 16V
D61 100V
R142 10K
R143 10M
TP33
TP34
PACK
LD
R144 5.1K
R145 5.1K
R146 5.1K
R147 5.1K
D62 100V
D63 100V


TS1
R96
0
EXTTS1
R99
0
RT1
103AT-2
C32
470pF
VSS
DDSG
R97
0
EXTDDSG
R100
0
RT2
103AT-2
C33
470pF
VSS
SPI_MOSI
R98
0
EXTSPI_MOSI
R101
0
RT3
103AT-2
C34
470pF
VSS
External thermistor interface
EXTDDSG 1
EXTDCHG 3
EXTDFETOFF 5
EXTCFETOFF 7
EXTSPI_MOSI 9
EXTALERT 11
EXTTS3 13
EXTTS2 15
EXTTS1 17
19
2
4
6
8
10
12
14
16
18
20
J9
VSS
TS2
R104
0
EXTTS2
R107
0
RT6
103AT-2
C37
470pF
VSS
DCHG
R102
0
EXTDCHG
R105
0
RT4
103AT-2
C35
470pF
VSS
DFETOFF
R103
0
EXTDFETOFF
R106
0
RT5
103AT-2
C36
470pF
VSS
TS3
R108
0
EXTTS3
R111
0
RT7
103AT-2
C38
470pF
VSS
ALERT
R109
0
EXTALERT
R112
0
RT8
103AT-2
C39
470pF
VSS
CFETOFF
R110
0
EXTCFETOFF
R113
0
RT9
103AT-2
C40
470pF
VSS